

WELL COMPLETION REPORT (SYNTHETIC)

OIL-DRK-004 - Dirok-004

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000044
Field / District	Dirok / Dibrugarh
Block	OALP-III-08
Basin	Assam-Arakan
Formation	Girujan (GRJ)
Spud / Completion	2013-03-12 / 2013-08-03
Measured / TVD	2147 m / 2119 m
Trajectory	Vertical; inc 1.0 deg; az 118.5 deg
Formation Interval	1278-1845 m
Lithology	Argillaceous sandstone and shale
Pore Pressure Reference	2388 psi
Fracture Gradient	13.91 ppg
Mud System	Glycol-PHPA WBM
NWIS Well Risk	Moderate (46.4/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	536	Surface
9 5/8 in casing	1245	Intermediate
7 in liner/casing	1889	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2013-06-21 1573 m	Torque Spike	Critical	Rotary speed fluctuated during constant pump rate. Mitigation: Ream and condition hole if torque remains elevated. Lesson: Use offset torque trend to set adaptive operating envelope.
2013-04-12 588 m	Stuck Pipe	High	Rotary movement deteriorated progressively. Mitigation: Improve hole cleaning, work pipe and back-ream within operating envelope. Lesson: Rising torque plus falling ROP should trigger hole-cleaning action.
2013-05-10 986 m	Stuck Pipe	High	Hookload became erratic during pipe movement. Mitigation: Improve hole cleaning, work pipe and back-ream within operating envelope. Lesson: Avoid prolonged static periods in high-drag interval.
2013-05-11 1007 m	Fishing Operation	High	Downhole tool remained unrecovered after freeing attempts. Mitigation: Use approved fishing program with controlled overpull and circulation. Lesson: Maintain detailed BHA inventory for rapid recovery planning.
2013-05-15 1054 m	Pressure Spike	High	Circulating pressure climbed without equivalent torque change. Mitigation: Condition hole and confirm pressure normalization. Lesson: Use depth-matched offset hydraulic baseline for early warning.
2013-07-05 1756 m	Mud Loss	High	Partial returns observed at shale shaker. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Prepare LCM package before approaching correlated offset-well loss depth.
2013-04-17 659 m	Cementing Issue	Medium	Cement density/returns trend indicated placement uncertainty. Mitigation: Verify cement placement using pressure and volume balance. Lesson: Plan conservative displacement rate in weak formations.

Completion Summary

The synthetic well reached 2147 m MD in the Girujan interval. The NWIS knowledge record links 15 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.